

CSA0903 – PROGRAMMING IN JAVA

CLASS ACTIVITY CO3

Sobitha D

192571046

1. Multiple catch statements

```
7
8 public class Main {
9     public static void main(String[] args) {
10         try {
11             int a = 10 / 0;
12             int arr[] = {1, 2};
13             System.out.println(arr[5]);
14         } catch (ArithmeticException e) {
15             System.out.println("Arithmetic Exception");
16         } catch (ArrayIndexOutOfBoundsException e) {
17             System.out.println("Array Index Exception");
18         }
19     }
20 }
```

input

Arithmetic Exception

...Program finished with exit code 0
Press ENTER to exit console.

2. StringIndexOutOfBoundsException

```
7
8 import java.util.Scanner;
9
10 public class Main {
11     public static void main(String[] args) {
12         Scanner sc = new Scanner(System.in);
13
14         try {
15             System.out.print("Enter a string: ");
16             String s = sc.nextLine();
17
18             System.out.print("Enter index: ");
19             int index = sc.nextInt();
20
21             System.out.println("Character: " + s.charAt(index));
22
23         } catch (StringIndexOutOfBoundsException e) {
24             System.out.println("Invalid string index.");
25         }
26     }
27 }
```

input

Enter a string: java
Enter index: 2
Character: v

...Program finished with exit code 0
Press ENTER to exit console.

3. finally

```
7
8 import java.util.Scanner;
9
10 public class Main{
11     public static void main(String[] args) {
12         Scanner sc = new Scanner(System.in);
13
14         try {
15             System.out.print("Enter first number: ");
16             int a = sc.nextInt();
17
18             System.out.print("Enter second number: ");
19             int b = sc.nextInt();
20
21             int result = a / b;
22             System.out.println("Result = " + result);
23
24         } catch (ArithmeticException e) {
25             System.out.println("Cannot divide by zero.");
26         } finally {
27             System.out.println("Finally block executed.");
28         }
29     }
30 }
```

input

Enter first number: 20
Enter second number: 15
Result = 1
Finally block executed.

...Program finished with exit code 0
Press ENTER to exit console.

4. throw (to check person is eligible to poll the vote)

```
7 *****/
8 import java.util.Scanner;
9
10 public class Main{
11     public static void main(String[] args) {
12         Scanner sc = new Scanner(System.in);
13
14         System.out.print("Enter age: ");
15         int age = sc.nextInt();
16
17         try {
18             if (age < 18)
19                 throw new Exception("Not Eligible to Vote");
20             else
21                 System.out.println("Eligible to Vote");
22         } catch (Exception e) {
23             System.out.println(e.getMessage());
24         }
25     }
26 }
```

input

Enter age: 16
Not Eligible to Vote

...Program finished with exit code 0
Press ENTER to exit console.

5. throws

```
8 import java.util.Scanner;
9
10 public class Main {
11
12     static void check(int age) throws Exception {
13         if (age < 18)
14             throw new Exception("Not Eligible to Vote");
15         else
16             System.out.println("Eligible to Vote");
17     }
18
19     public static void main(String[] args) {
20         Scanner sc = new Scanner(System.in);
21
22         System.out.print("Enter age: ");
23         int age = sc.nextInt();
24
25         try {
26             check(age);
27         } catch (Exception e) {
28             System.out.println(e.getMessage());
29         }
30     }
31 }
```

input

Enter age: 20
Eligible to Vote

...Program finished with exit code 0
Press ENTER to exit console.